$\forall t, r \in \mathbb{N}$, t & r don't affect the cov

$$cov(S_t, S_r) = cov\left(\sum_{i=1}^{t} X_i, \sum_{j=1}^{r} X_j\right)$$

$$= \sum_{i=1}^{t} \sum_{j=1}^{r} cov(X_i, X_j)$$

knowing that $cov(X_i, X_j) = \begin{cases} \sigma^2, & i = j \\ 0, & \text{else} \end{cases}$

$$cov(S_t, S_r) = \begin{cases} t\,\sigma^2, & \text{if } t \geq s \\ r\,\sigma^2, & \text{else} \end{cases}$$

finally $\boxed{cov(S_t, S_r) = \min(t, r)\,\sigma^2}$

b)

$\nabla S_t = S_t - S_{t-1}$ by using the general form of S_t we have

$$\nabla S_t = \mu + X_t$$

$\{\nabla S_t\}$ is stationary because $\{X_t\}$ is stationary

$$E[\nabla S_t] = E[\mu + X_t] = \mu + \underbrace{E[X_t]}_{0}$$

$$\boxed{E[\nabla S_t] = \mu}$$